VATSALKUMAR ARVINDKUMAR BHIMANI

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EDUCATION

California State University – Los Angeles **Aug 2024 – May 2026**

- Master of Science in Computer Science ***GPA: 3.7***

Gujarat Technological University, Ahmedabad, India **Jun 2020 – Jun 2024**

- Bachelor of Engineering in Computer Science & Engineering (AI/ML) ***GPA: 3.42***

SKILLS

Programming & Query Languages: Python, SQL (Joins, Aggregations, Subqueries, CTEs, Query Optimization)

Data Analysis & Statistical Techniques: Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, A/B Testing, Regression Analysis, Time Series Analysis, Data Validation

Data Visualization & Business Intelligence: Power BI, Tableau, Excel (Pivot Tables, VLOOKUP, XLOOKUP), KPI Reporting, Dashboard Development, Data Storytelling

Data Engineering & Data Management: ETL Pipelines, Data Cleaning, Data Transformation, Data Wrangling, Data Modeling (Schema Design, Normalization), Data Warehousing, Data Quality Assessment

Databases: PostgreSQL, MongoDB, SQLite

Tools: Git, Jupyter Notebook, Streamlit

EXPERIENCE

SculptSoft Pvt. Ltd. - Ahmedabad, Gujarat, India **Jan 2024 – Apr 2024**

Machine Learning Intern

Tools & Technologies*: Python (Pandas, NumPy, Scikit-learn), TensorFlow, Streamlit, SQLite, Passlib*

- Conducted exploratory data analysis (EDA) and statistical analysis on structured datasets to identify patterns and improve model performance by 14.5%, enhancing predictive reliability, while collaborating with cross-functional teams to deliver actionable insights improving decision-making efficiency by 27.5%.
- Built data preprocessing and validation pipelines using Python (Pandas, NumPy), reducing data inconsistencies by 32% and improving data quality, and developed 3+ Streamlit dashboards to visualize model outputs and performance metrics.
- Implemented secure authentication using SQLite and Passlib with encrypted credential storage and role-based access.

PROJECTS

Healthcare Comorbidity & Patient Outcome Analysis **Feb 2026 - March 2026**

Tableau, Excel/Google Sheets, GitHub, Tableau Public

- Developed interactive Tableau dashboards leveraging calculated fields to analyze 101K+ hospital encounters, identifying patterns in readmissions, length of stay, and discharge outcomes across 8 comorbidity segments.
- Performed data cleaning and transformation in Excel/Google Sheets, standardizing demographics, diagnosis, and treatment variables, and engineered comorbidity segmentation using ICD codes for Diabetes, Heart Disease, and Cancer combinations, improving analysis efficiency by ~40% across 8 groups.
- Designed multi-visual analytics (heatmaps, bubble charts, discharge distribution) to uncover trends in average hospital stay (~4.4 days) and treatment outcomes, supporting data-driven healthcare insights.

CricIQ - Cricket Analytics Platform **Aug 2024 - Dec 2024**

Python, PostgreSQL, MongoDB, Power BI

- Architected a cricket analytics platform processing 100K+ structured records using PostgreSQL and MongoDB for player and match performance analysis.
- Built ETL workflows in Python (Pandas) to extract, transform, and load 100K+ records into relational and NoSQL databases, improving data consistency.
- Designed 4+ BI dashboards using Power BI to visualize player KPIs and match metrics, and engineered a SQL-based performance tracking system with advanced filtering, reducing manual analysis time by 40%.

Predicting Stocks Prices **Jan 2024 - Apr 2024**

Python, TensorFlow, Streamlit, SQLite

- Retrieved and analyzed 10 years (~15K+ records) of stock data using yfinance, performed time-series EDA, and cleaned data with Python (Pandas, NumPy) to improve forecasting accuracy.
- Trained and optimized an LSTM model (TensorFlow) achieving 85% accuracy and reducing forecasting error by 20%.
- Developed a Streamlit dashboard for 100+ stock symbols with interactive filtering and implemented authentication using SQLite with encrypted password hashing.